

Lattice Resonance in a Connected Algebraic Action: Complete Quotients and Continuous Fixed Strata

HCS-C388 source-theorem and reproducibility package

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Abstract

We reconstruct every finite-index fixed group of the connected algebraic action defined by the integral relation $1 + u + v$. A canonical Hermite lattice presentation produces an exact integer convolution matrix, whose Smith factors determine the complete compact-group type. There are exactly two possible Fourier zero modes. They occur precisely when the acting lattice lies in one fixed index-three subgroup; otherwise the fixed group is finite. At resonance the fixed group has a two-dimensional torus factor. We give the full quotient conventions, including coincident neighbors and skew lattices, and distinguish finite cardinality from connected components. This is a source-local reconstruction with direct classical ownership, not a claim to have discovered the resonance condition. No scalar prime-time clock or target determinant is supplied by the quotient atlas.

中文摘要

本文重建由整数关系定义的紧连通二维代数作用的全部有限指数固定群。
规范埃尔米特格表示给出精确整数卷积矩阵，其史密斯不变量确定固定群的完整紧群类型。
傅里叶分解中只有两个可能零模；当且仅当作用子格包含于指定的指数三子格时出现共振，固定群因而含二维环面。
非共振情形的固定群有限。本文明确斜子格与重合邻点约定，并区分固定点总数和连通分支数。
这些源系统结论具有直接经典文献归属，不被包装成新发现，也没有提供素数飞行时间或目标行列式。

Keywords: algebraic action; lattice resonance; Smith form; integer covolume; Mahler measure; orbit counting
关键词：代数作用；子格共振；史密斯形；整数格体积；马勒测度；轨道计数

1 One relation, two different kinds of fixed groups

A Fourier zero is not merely an eigenvalue to omit from a determinant. For a compact connected algebraic action, it can create a continuous family of periodic states. The relation studied here is elementary:

$$X = \{x \in \mathbb{T}^{\mathbb{Z}^2} : x_{i,j} + x_{i+1,j} + x_{i,j+1} = 0\}, \quad (\alpha^{(a,b)}x)_{i,j} = x_{i+a,j+b}, \quad \mathbb{T} = \mathbb{R}/\mathbb{Z}. \quad (1)$$

The challenge addressed by this reconstruction is to retain the integral lattice while passing between fixed groups, Fourier modes and finite component counts. Real diagonalization alone discards that information.

Every acting sublattice has a unique column Hermite normal form

$$\Lambda = \langle (a,0), (b,c) \rangle, \quad a, c > 0, \quad 0 \leq b < a, \quad N = [\mathbb{Z}^2 : \Lambda] = ac. \quad (2)$$

We write Fix_Λ for points fixed by every element of Λ , not for points whose stabilizer is exactly Λ . The central source dichotomy is finite versus a finite union of two-tori.

Classical ownership and the precise scope. Lind, Schmidt and Ward identify the entropy of principal algebraic actions with Mahler measure, and display this exact polynomial in their introduction [1]. Smyth owns its special-value evaluation [2]. Lind, Schmidt and Verbitskiy give the exact Hermite congruence and the finite-versus-two-torus distinction in Example 3.2 of the accessed source [3]. None of those facts is presented here as a literature discovery. The package supplies an explicit all-lattice reconstruction, complete integer certificates and a careful distinction between the quantities that an orbit-count construction needs.

2 Connected source and exact quotient matrices

The relation in (1) is closed in a product of circles, so X is a compact metrizable abelian group. Coordinate shifts are automorphisms. Its discrete character module is

$$\widehat{X} = \mathbb{Z}[u^{\pm 1}, v^{\pm 1}] / (1 + u + v) \simeq \mathbb{Z}[u^{\pm 1}, (1 + u)^{-1}]. \quad (3)$$

The right-hand ring embeds in $\mathbb{Q}(u)$ and is additively torsion-free. The duality criterion for compact abelian groups therefore makes X connected. This is not a finite-characteristic three-dot shift: its coordinates are circles, and the scalar ring is the integers.

Let $G = \mathbb{Z}^2/\Lambda$. Choose representatives (i, j) with $0 \leq i < a$, $0 \leq j < c$ in lexicographic order. To reduce any integer pair, write $j = qc + r$, $0 \leq r < c$, and replace it by $((i - qb) \bmod a, r)$. Let S_1, S_2 be the permutation matrices $(S_k x)_g = x_{g+e_k}$. They commute and are orthogonal. Define

$$A_\Lambda = I + S_1 + S_2, \quad \text{Fix}_\Lambda = \ker(A_\Lambda : \mathbb{T}^N \longrightarrow \mathbb{T}^N). \quad (4)$$

If two neighbors coincide, their contributions add. In particular, at index one the matrix is $[3]$, not $[1]$. Equation (4) is exact because a Λ -periodic array is determined by its quotient coordinates and the defining relation must hold at every quotient site.

Theorem 1 (Complete effective fixed-group presentation). *Suppose integer unimodular matrices U, V put A_Λ into Smith form*

$$UA_\Lambda V = \text{diag}(d_1, \dots, d_r, 0, \dots, 0), \quad 0 < d_1 \mid \dots \mid d_r.$$

Then

$$\text{Fix}_\Lambda \simeq \mathbb{T}^{N-r} \times \prod_{j=1}^r \mathbb{Z}/d_j\mathbb{Z}. \quad (5)$$

This classifies every finite-index lattice fixed group by an explicit integer algorithm. The splitting need not be canonical.

Proof. The matrices U, V induce torus automorphisms since their inverse matrices are integral. A diagonal map $t \mapsto d_j t$ on a circle has kernel $\mathbb{Z}/d_j\mathbb{Z}$, whereas a zero diagonal entry leaves a whole circle free. Applying these facts coordinatewise proves (5). Smith normal form terminates by exact integer Euclidean row and column operations; no numerical eigenvalue threshold enters the statement. $\square$

3 All Hermite lattices and the two resonant modes

Write $\omega = \exp(2\pi i/3)$ and $K = \{(i, j) \in \mathbb{Z}^2 : i - j \equiv 0 \pmod{3}\}$. Finite characters of G diagonalize (4); their eigenvalues are $1 + \chi(e_1) + \chi(e_2)$, including every character multiplicity. For unit complex numbers z, w , $1 + z + w = 0$ implies $1 = |1 + z|^2 = 2 + 2 \operatorname{Re} z$. Hence

$$(z, w) \in \{(\omega, \omega^2), (\omega^2, \omega)\}. \quad (6)$$

The first character descends through G exactly when $\Lambda \subset K$; the conjugate has the identical condition.

Theorem 2 (The complete resonance test). *For (2), resonance is equivalent to*

$$3 \mid a, \quad b \equiv c \pmod{3}. \quad (7)$$

At resonance the nullity of A_Λ is exactly two and Fix_Λ is a finite union of two-tori. Otherwise it is finite, with cardinality

$$|\text{Fix}_\Lambda| = |\det A_\Lambda| = \prod_{\chi \in \widehat{G}} |1 + \chi(e_1) + \chi(e_2)|. \quad (8)$$

Proof. The character $(i, j) \mapsto \omega^{i-j}$ is trivial on the two lattice generators precisely under (7). Equation (6) excludes all other zero modes. Fourier diagonalization gives the nullity; Theorem 1 then proves the compact-group assertion. In the invertible case, the number of kernel points is the determinant's absolute value. These arguments reproduce the classical test in [3, Example 3.2] with the current forward-shift convention. $\square$

Index does not decide resonance. For $(a, c) = (3, 1)$, the shear $b = 1$ is resonant and $b = 0, 2$ are not. Thus a census based only on the index loses the continuous fixed stratum. For rectangular lattices $b = 0$, the criterion becomes $3 \mid a$ and $3 \mid c$. Prime indices, composite indices, and neighboring shears use the same source rule; no rational-prime labels are attached to the orbits.

4 Evidence, literature boundary and conclusion

The canonical producer retains every quotient matrix and its integer Smith witnesses. An independent checker imports neither that producer nor a symbolic algebra package: direct multiplication checks the factorization, Bareiss elimination verifies unimodularity, and Faddeev–LeVerrier verifies all characteristic coefficients. A separate library Smith calculation and character-by-character Fourier product give additional controls. The finite

grid contains 142 HNF lattices: twenty resonant and 122 nonresonant. It includes every index at most twelve and all shears for $(a, c) = (3, 6), (6, 3), (6, 6)$. These finite receipts test the implementation; the universal theorem is proved independently. Repaired-hash attacks specifically distinguish a pseudodeterminant from a component count, and raw/type-locked evaluation checks reject unknown fields and boolean-to-integer drift.

The direct literature owners must remain visible. Principal-action entropy is an instance of [1]; the exact source resonance is [3, Example 3.2]; the special value belongs to [2]. General periodic-component growth results in [3, Theorem 1.2] are not replaced by a claim about ordinary point-cardinality zeta. The present reconstruction is owner-heavy and is not a novelty certificate.

The strict Route-A assessment is weak arithmetic relation, weak primitive layer, and failure of the target determinant and target analytic layers. Native commuting Haar Koopman unitaries give only a formal quantum hint; no privileged scalar generator is constructed. No target arithmetic local data, Euler factors, root number, automorphy, target-zero match or Hilbert–Polya operator is claimed. Route B remains disabled. The scope literal is NO_BAD_EULER_OR_ROOT_NUMBER.

References

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Round zero: complete quotient presentations and the resonance test.